

Exercise No. 2

ADVANCED ENCRYPTION STANDARD

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3.1

Plain text:

Store leftover pancakes in an airtight container in the fridge for about a week. Refrain from adding toppings (such as syrup) until right before you serve them so the pancakes don't get soggy. To reheat pancakes, you have a few options: warm a batch in a 350 degrees F oven (covered) for 8 to 10 minutes, microwave individual pancakes in 20 to 30 second bursts with a damp paper towel, or reheat them in a skillet over medium-low heat for 30–60 seconds per side until warmed through.

Key:

Histogram of plain text:

Histogram of encryption:

Conclusions:

AES exhibits excellent "confusion" and "diffusion." The uniform histogram proves that the algorithm successfully destroys any statistical patterns present in the plaintext, rendering frequency analysis attacks completely useless. Decrypting the text with the same key will perfectly restore the original uneven histogram.

3.2

Changes:

- State Matrices:
- Round Keys:

After how many rounds, the outcome is similar to random noise?:

Looks like after 2-3.

Why do we continue up to 10 times?:

To ensure complete security. While 3 rounds might look visually random to a human, mathematical correlations still exist. Ten rounds guarantee that every single bit of the output ciphertext is a complex, non-linear function of every single bit of the plaintext and the key, making cryptanalysis computationally infeasible.

3.3

Plain text:

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Key:

Encryption:

Decryption with Key :

Results:

Decrypted text is completely unreadable.

Observed effect – the Avalanche Effect (specifically, the Strict Avalanche Criterion). In a well-designed cipher like AES, a change in just 1 bit of the key (or plaintext) results in approximately a 50% change in the resulting output bits. Because the decryption key schedule diverges entirely from the correct key schedule after that 1-bit flip, the algorithm outputs random noise instead of the original text.

3.4

Plain text:

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Key:

Encryption:

Decryption of slightly changed text:

Decryption of massively changed text:

Results:

AES operates on 16-byte blocks. Any error within a ciphertext block ruins the entire corresponding plaintext block upon decryption. Introducing a whole series of changes will permanently corrupt every 16-byte block touched by those changes.

3.5

Images: colors.bmp, car.bmp

Comparison:

- ECB: The encrypted image still displays the clear visual outline, shapes, and color groupings of the original car or colors. Identical plaintext blocks of color are encrypted into identical ciphertext blocks.
- CBC: The encrypted image looks like pure static or TV snow. No visual patterns from the original image remain.
- PCBC: Similar to CBC, the image looks like completely random static noise.

Conclusions:

ECB mode is highly insecure for data with repeating patterns (like image headers or solid color blocks) because it does not hide the underlying data structure. CBC and PCBC modes XOR the current block with the previous ciphertext block, effectively hiding all repetitive patterns.

3.6

Results:

- ECB: The bottom-left corner of the image has a small block of scrambled, colorful static. The entire rest of the image is perfect.
- CBC: The bottom-left corner is scrambled, and the block immediately following it has corrupted pixels. The rest of the image recovers and displays perfectly.
- PCBC: The bottom-left corner is scrambled, and the entire rest of the image is completely destroyed.

Conclusions:

PCBC mode has an infinite error propagation property. An error in one block cascades forward and corrupts every single block that comes after it during decryption.

3.7

Time for 13 bits key:

Time for 12 bits key:

Time for 11 bits key:

Results:

It takes multiple orders of magnitude more time to brute force AES than the old DES.

3.8

Message:

Decryption time for 128 bit key:

Decryption time for 192 bit key:

Decryption time for 256 bit key:

Results:

Attacking AES-192 takes 2^{64} times longer than attacking AES-128.

Attacking AES-256 takes 2^{128} times longer than attacking AES-128.

3.9

Advantages:

- Highly Secure: Resistant to all known practical attacks (differential and linear cryptanalysis).
- Efficient and Fast: Performs extremely well in both software and hardware across various platforms.
- Flexible Key Sizes: Offers 128, 192, and 256-bit lengths depending on security needs.
- Standardized: Globally accepted, thoroughly peer-reviewed, and standardized by NIST.

Disadvantages:

- Symmetric Key Issue: Both parties must safely exchange the key beforehand; if the key is intercepted, the system falls.
- Implementation Risks: If programmed poorly, AES is vulnerable to side-channel attacks (like timing or power analysis attacks).
- Mode Dependencies: Relying on weak modes of operation (like ECB) undermines the entire security of the algorithm.